

# The Theory of True Services

Soumadeep Ghosh with ChatGPT (OpenAI)

Kolkata, India

## Abstract

This paper develops a formal theory of True Services, defined as economic activities that are non-standardizable, non-replicable, and possess zero elasticity of substitution with respect to all other services and products. We investigate implications for economics, finance, political science, geopolitics, warfare economics, artificial intelligence, algorithmic systems, and social organization. The theory suggests that certain forms of value originate not from scalable production but from unique relational, contextual, and informational interactions.

The paper ends with “The End”

## Contents

|                                        |          |
|----------------------------------------|----------|
| <b>1 Introduction</b>                  | <b>2</b> |
| <b>2 Definitions</b>                   | <b>2</b> |
| <b>3 Utility Theory</b>                | <b>3</b> |
| <b>4 Elasticity Structure</b>          | <b>3</b> |
| <b>5 Pricing Theory</b>                | <b>4</b> |
| 5.1 Uniqueness Premium . . . . .       | 4        |
| <b>6 Financial Interpretation</b>      | <b>4</b> |
| <b>7 Information Theory</b>            | <b>4</b> |
| <b>8 Political Science</b>             | <b>5</b> |
| <b>9 International Relations</b>       | <b>5</b> |
| <b>10 Warfare Economics</b>            | <b>5</b> |
| <b>11 Artificial Intelligence</b>      | <b>6</b> |
| 11.1 AI Boundary . . . . .             | 6        |
| <b>12 Machine Learning Formulation</b> | <b>6</b> |
| <b>13 Algorithmic Identification</b>   | <b>7</b> |
| 13.1 Pseudo-Code . . . . .             | 7        |
| <b>14 Statistical Model</b>            | <b>7</b> |
| <b>15 Comparative Classification</b>   | <b>8</b> |

|                                      |          |
|--------------------------------------|----------|
| <b>16 Conceptual Diagram</b>         | <b>8</b> |
| <b>17 Philosophical Implications</b> | <b>8</b> |
| <b>18 Main Result</b>                | <b>8</b> |
| <b>19 Conclusion</b>                 | <b>9</b> |

## 1 Introduction

Classical economics frequently models goods and services as substitutable objects occupying a common utility space. This paper studies a limiting case:

A True Service is a service whose value depends intrinsically upon its unique realization and therefore cannot be substituted by any other service or product.

Examples may include:

- trust-based negotiations,
- unique diplomatic interventions,
- singular military decisions,
- irreplaceable mentoring relationships,
- context-specific professional judgments,
- historically unique acts.

## 2 Definitions

**Definition 1** (Product). *A product is a collection of economically identical service realizations:*

$$G = \{s_1, s_2, \dots, s_n\}$$

*such that*

$$s_i \cong s_j$$

*for all  $i, j$ .*

**Definition 2** (True Service). *A True Service  $T$  satisfies:*

1.  *$T$  is not a product.*
2.  *$T$  is not economically identical to any other service.*
- 3.

$$\sigma(T_i, T_j) = 0$$

*for all  $i \neq j$ .*

- 4.

$$\sigma(T, G) = 0$$

*for every product  $G$ .*

*where  $\sigma$  denotes elasticity of substitution.*

### 3 Utility Theory

Let utility be

$$U = U(T_1, T_2, \dots, T_n, G_1, \dots, G_m).$$

For products, standard CES utility permits substitution:

$$U_{CES} = \left( \sum_i \alpha_i G_i^\rho \right)^{1/\rho}.$$

True Services violate CES assumptions.

Instead:

$$U = \sum_i u_i(T_i) + \sum_j v_j(G_j).$$

The marginal contribution of each True Service remains independent:

$$\frac{\partial U}{\partial T_i} \neq 0$$

even when all other services exist.

### 4 Elasticity Structure

For CES systems,

$$\sigma = \frac{1}{1 - \rho}.$$

The theory imposes

$$\sigma = 0.$$

This corresponds to the limiting case

$$\rho \rightarrow -\infty.$$

Thus True Services occupy distinct utility dimensions.

**Theorem 1.** *No True Service can be replaced by any combination of products.*

*Proof.* Suppose

$$T = f(G_1, \dots, G_n).$$

Then a marginal increase in products reproduces the service.

This implies

$$\sigma(T, G_k) > 0.$$

Contradiction.

Therefore

$$\sigma(T, G_k) = 0$$

precludes replacement.

□

## 5 Pricing Theory

Let

- $r_f$  be the risk-free rate,
- $p_{ts}$  be the True-Service premium.

Define

$$P_0 = \frac{P_1}{1 + r_f + p_{ts}}.$$

Unlike conventional assets,  $p_{ts}$  emerges from uniqueness rather than risk.

### 5.1 Uniqueness Premium

Let

$$UQ(T) \in [0, 1]$$

measure uniqueness.

Then

$$p_{ts} = \alpha UQ(T).$$

As uniqueness increases:

$$\frac{\partial p_{ts}}{\partial UQ} > 0.$$

## 6 Financial Interpretation

Traditional asset pricing:

$$E(R_i) = r_f + \beta_i(E(R_M) - r_f).$$

For True Services:

$$\beta_T \rightarrow 0.$$

However:

$$p_{ts} \rightarrow \text{positive}.$$

Thus value is not explained by systematic risk.

## 7 Information Theory

Let

$$I(T)$$

denote information contained in a True Service.

If replication is impossible:

$$K(T) = \text{Kolmogorov Complexity}$$

is maximal relative to feasible descriptions.  
Therefore:

$$I(T) > I(G)$$

for equivalent product categories.

## 8 Political Science

Political legitimacy often depends upon non-replicable acts.

Examples include:

- treaty negotiations,
- constitutional settlements,
- crisis leadership.

A unique political intervention may have:

$$\sigma(T_{political}, G) = 0.$$

Mass bureaucratic processes cannot substitute for it.

## 9 International Relations

Diplomatic outcomes frequently depend upon unique interpersonal relationships.

Let

$$D = f(T_1, T_2, \dots, T_n).$$

Then international stability depends partly on irreplaceable services.

Examples:

- back-channel diplomacy,
- ceasefire mediation,
- confidence-building measures.

## 10 Warfare Economics

Military systems combine products and True Services.

Products:

- ammunition,
- vehicles,
- fuel.

True Services:

- battlefield judgment,
- strategic innovation,

- deception planning.

Military output:

$$M = F(G, T).$$

Even if

$$G \rightarrow \infty,$$

absence of key True Services may imply

$$M \rightarrow 0.$$

## 11 Artificial Intelligence

AI excels at scalable service production.

Let

$$A = \text{AI capability.}$$

Then product-like services satisfy

$$\lim_{A \rightarrow \infty} \text{Cost}(G) = 0.$$

However:

$$\text{Cost}(T) \not\rightarrow 0.$$

if uniqueness is essential.

### 11.1 AI Boundary

Define

$$B(T) = P(T \text{ is replicable by AI}).$$

Then:

$$B(T) = 0$$

for ideal True Services.

## 12 Machine Learning Formulation

Let

$$X = x_1, \dots, x_n$$

be training data.

A model learns:

$$f_\theta : X \rightarrow Y.$$

If a service is unique and unrepeatable:

$$P(x_i = x_j) = 0.$$

Generalization becomes impossible.

Theoretical limit:

Accuracy  $\rightarrow$  random baseline.

## 13 Algorithmic Identification

### 13.1 Pseudo-Code

INPUT: Service S

IF S is standardized THEN

RETURN Product

END IF

Compute substitution elasticity sigma

IF sigma > 0 THEN

RETURN Non-True-Service

END IF

Compute uniqueness score UQ

IF UQ exceeds threshold THEN

RETURN True Service

ELSE

RETURN Non-True-Service

END IF

## 14 Statistical Model

Let

$$Y_i = 1$$

if service  $i$  is classified as True.

Logistic specification:

$$P(Y_i = 1) = \frac{1}{1 + \exp(-z_i)}$$

where

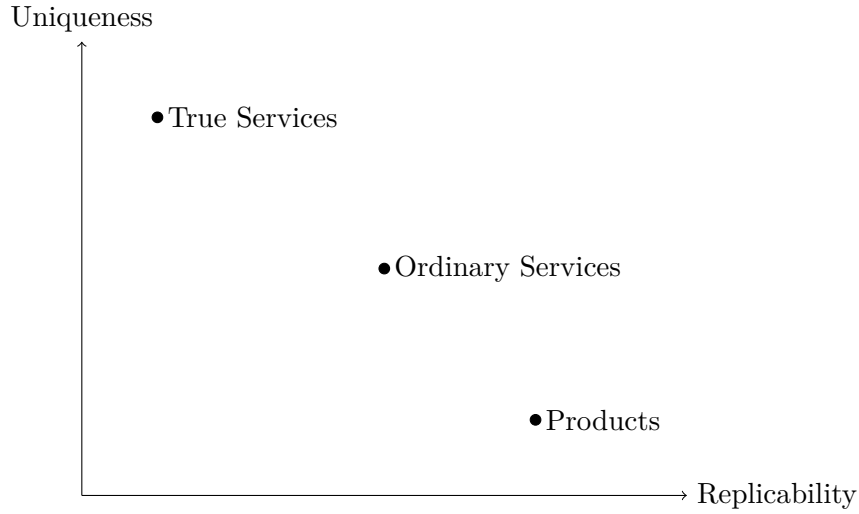
$$z_i = \beta_0 + \beta_1 UQ_i - \beta_2 \sigma_i.$$

## 15 Comparative Classification

| Type             | Replicable | Substitutable | Scalable  |
|------------------|------------|---------------|-----------|
| Product          | Yes        | Yes           | Yes       |
| Ordinary Service | Partially  | Partially     | Partially |
| True Service     | No         | No            | No        |

Table 1: Economic classification.

## 16 Conceptual Diagram



## 17 Philosophical Implications

The theory implies that value may arise from singularity rather than reproducibility.

Classical industrial civilization prioritizes:

Scale  $\rightarrow$  Efficiency.

The Theory of True Services suggests:

Uniqueness  $\rightarrow$  Irreplaceable Value.

This shifts attention from production functions toward relational ontology.

## 18 Main Result

**Theorem 2.** *Suppose a service  $T$  satisfies:*

- 1. non-replicability,*
- 2. non-standardizability,*
- 3. zero substitution elasticity.*

*Then no finite combination of products can reproduce  $T$ .*



*Proof.* Assume reproduction exists:

$$T = f(G_1, \dots, G_n).$$

Then at least one product contributes positively:

$$\frac{\partial T}{\partial G_i} > 0.$$

Hence:

$$\sigma(T, G_i) > 0.$$

Contradiction.

Therefore reproduction is impossible.

□

## 19 Conclusion

The Theory of True Services introduces a category of economic activity characterized by uniqueness, non-replicability, and zero substitutability. Such services differ fundamentally from products and standard services. The framework provides implications for economics, finance, political systems, warfare, artificial intelligence, and information theory. Future work may develop empirical measures of uniqueness and estimate True-Service premia across institutional environments.

## References

- [1] Arrow, K. J. *The Limits of Organization*. W. W. Norton.
- [2] Becker, G. S. *The Economic Approach to Human Behavior*. University of Chicago Press.
- [3] Hayek, F. A. *The Use of Knowledge in Society*. American Economic Review.
- [4] Marshall, A. *Principles of Economics*. Macmillan.
- [5] Samuelson, P. A. *Foundations of Economic Analysis*. Harvard University Press.
- [6] Shannon, C. E. A Mathematical Theory of Communication. Bell System Technical Journal.
- [7] Turing, A. M. Computing Machinery and Intelligence. Mind.
- [8] von Neumann, J. and Morgenstern, O. *Theory of Games and Economic Behavior*. Princeton University Press.
- [9] Wiener, N. *Cybernetics*. MIT Press.

## **Glossary**

### **True Service**

A unique service possessing zero substitution elasticity.

### **Product**

A collection of economically identical service realizations.

### **True-Service Premium**

Additional value arising from uniqueness.

### **Elasticity of Substitution**

Ability of one good or service to replace another.

### **Uniqueness Score**

Quantitative measure of non-replicability.

### **Replicability**

Degree to which a service can be reproduced.

### **Strategic Service**

A True Service affecting military, political, or diplomatic outcomes.

## **The End**